Initial Discussion Post – Industry 4.0 & 5.0 in the Luxury Fashion Space

The 4th Industrial Revolution

Several industrial revolutions have occurred in history, from the introduction of steam and water to produce mechanical production equipment, to using electricity to aid manufacturing, before the introduction of automation using information technology (IT). The fourth and latest revolution, also referred to Industry 4.0, is a period where physical, digital, and biological systems are merged using technologies such as artificial intelligence (AI), robotics, and big data (Schwab, 2016). Application of these technologies is vast, ranging from the development of mobile apps to order various goods without having to step into a physical store, to developing machine learning (ML) to drive marketing activities.

Fashion brands have leveraged technological advancements such as AI and big data analytics to accurately forecast product demand and subsequent production, as well as issue targeted promotions to customers based on their spending habits (Jin and Shin, 2021). Data from customers is instrumental in the development of AI and predictive models, presenting significant risks and vulnerabilities.

Major luxury fashion houses were targeted by cyber criminals in 2025, stealing the personal data of millions of customers. The criminals, who refer to themselves as ‘Shiny Hunters’, stole the names, email addresses, phone numbers, addresses, and spend totals of over seven million Gucci, Alexander McQueen, and Balenciaga customers. Reputational damage was inflicted as a result and customers reported lower levels of trust after the attack, especially because the hacker could pass on their details to secondary scammers, who would issue phishing attacks. Given the high value purchases that customers usually make, hackers would likely attempt to gain several thousands of pounds if successful. This directly affected sales and revenue. Another financial risk lies with the ransom which Shiny Hunters demanded that Kering, the parent company of these brands, paid (Tidy, 2025).

Industry 5.0 seeks to incorporate a human-centric and sustainable approach when using technologies (Metcalf, 2024). Instead of purely using AI when forecasting trends or to speed up production, it can be used as a reinforcement measure alongside human intelligence to improve cyber threat detection and analysis, which reduces response times to cyber incidents in the event of an attack (Chaudhuri *et al.*, 2025). Overreliance on modern technologies without consideration for the role of humans is not a sustainable approach long-term, and a shift to a well-rounded partnership ethos will aid the resilience and eventually prosperity of businesses within the fashion industry.

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Peer Response #1:

Williams (2026) highlights a major 2025 breach where the *ShinyHunters* gang stole millions of customers' details from Gucci and related brands (Hummel, 2025). Preventing such attacks requires comprehensive cybersecurity hygiene. First, retailers must know and secure all data assets. Professor Li (2025) warns that patchwork legacy systems and weak third-party integrations leave retailers vulnerable. In response, companies should conduct thorough vendor security audits and enforce strict access controls. For example, limiting who can access customer databases and encrypting personal data significantly reduces impact if a breach occurs.

Second, continuous monitoring is essential. Cybersecurity experts advise running automated leak-detection tools on data repositories: Traynor (2024) notes that organisations should inventory all sensitive data and scan for any leaks, including monitoring dark-web forums for stolen credentials. Third, employee training is critical. Many breaches start with phishing or human error; as Kumar (2025) points out, staff should be trained to recognise suspicious emails and follow security protocols.

By combining technical controls (encryption, segmentation, leak detection) with human-centred measures (training, incident plans), organisations can greatly reduce risk. These steps, all aligned with Li's (2025) call for securing integrations and Kumar's (2025) emphasis on awareness would help prevent the kind of data theft Williams (2026) describes.

References

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Summary Post:

In my initial post, I discussed the progression of the several industrial revolutions, from steam and electricity through to Industry 4.0, where AI, robotics and big data now merge physical and digital systems. Fashion brands use these tools to forecast demand and target promotions, but this reliance on customer data creates serious vulnerabilities. I detailed the 2025 ShinyHunters attack, in which millions of records belonging to Gucci, Alexander McQueen and Balenciaga customers were stolen, damaging trust and exposing victims to phishing and financial harm, while Kering also faced a ransom demand (Tidy, 2025). Industry 5.0 offers a way forward, pairing AI with human oversight to strengthen cyber threat detection rather than depending on automation alone.

My peer Preh built on this analysis, focusing on prevention. They argued that retailers must first map and secure their data assets, since outdated systems and poor third-party integrations are common weak points. Continuous monitoring, including automated leak detection and dark-web scanning, is presented as a second essential layer, alongside encryption and access controls. Their third point centres on staff training, since human error and phishing remain leading causes of breaches. Combining technical safeguards with awareness-building offers the most realistic route to reducing risk across the sector.

References

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